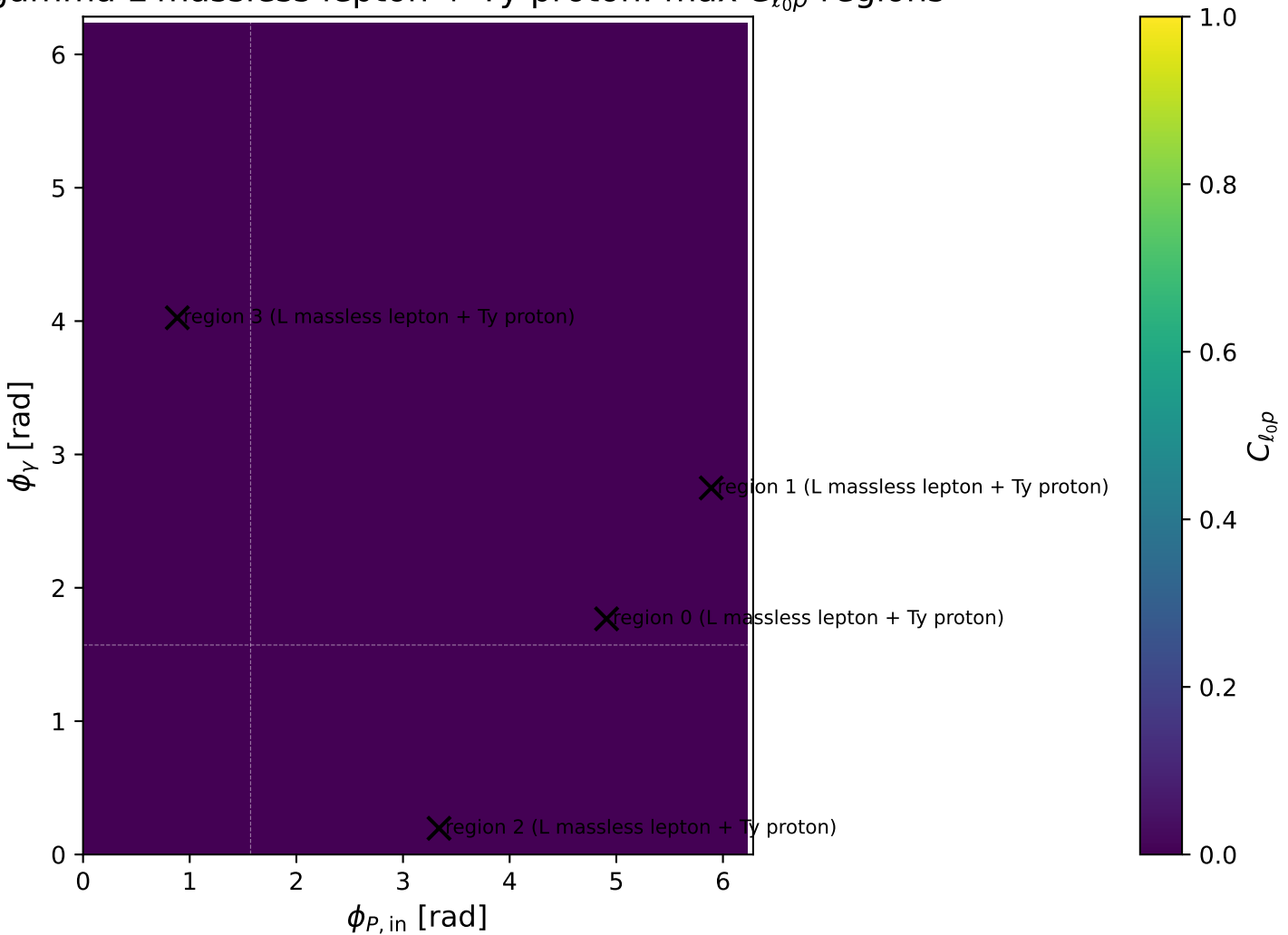
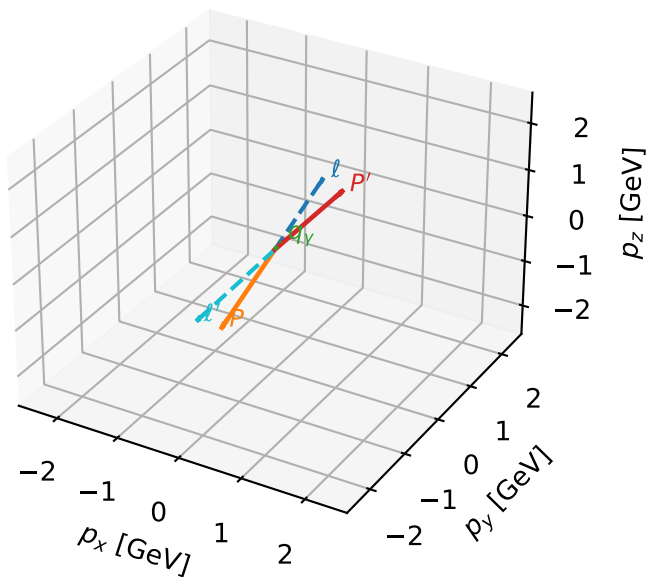


gamma L massless lepton + Ty proton: max  $C_{\ell_0 p}$  regions



# L massless lepton + Ty proton characteristic max $C_{\ell_0 p}$ configuration

3D momenta



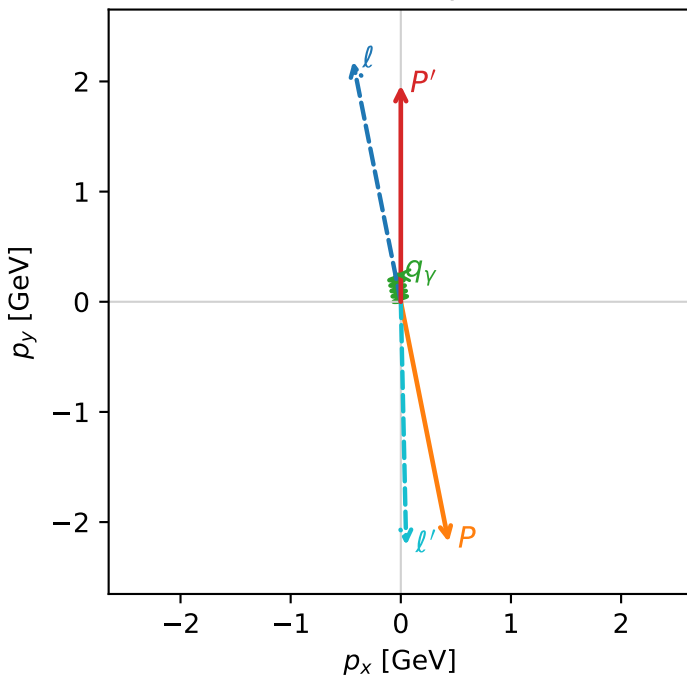
c\_ep\_low egamma\_lty\_region\_0 C\_{\ell\_0 p}=1.84225e-10  
 region: low\_Egamma\_LTy\_region\_0 final-pair delta\_xy=3.11953 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_{t_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=1.9652$   
 $\theta_{in}=1.57079$ ,  $\phi_{\ell}=1.76715$ ,  $\phi_p=4.90874$   
 $E_{\gamma}=0.25$ ,  $\phi_{\gamma}=1.76715$   
 $Q^2=19.5232$ ,  $x_B=0.993635$ ,  $t=-17.329$ ,  $W^2=1.00491$ ,  $y=0.952134$   
 $\theta(\ell', \gamma)=170.014$  deg

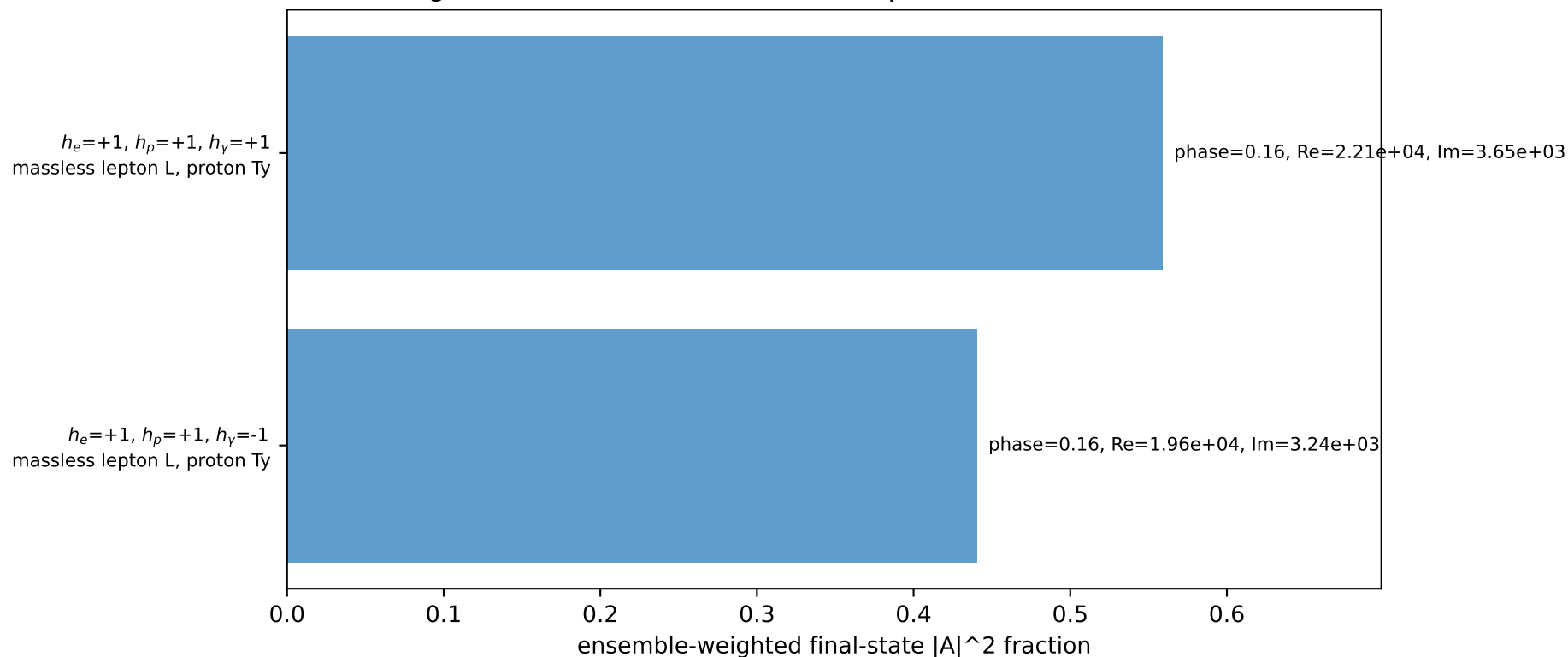
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

|              |             |                |                |                |
|--------------|-------------|----------------|----------------|----------------|
| $\ell$       | $E= 2.2244$ | $p_x= -0.4340$ | $p_y= 2.1817$  | $p_z= -0.0000$ |
| $P$          | $E= 2.4141$ | $p_x= 0.4340$  | $p_y= -2.1817$ | $p_z= 0.0000$  |
| $\ell'$      | $E= 2.2109$ | $p_x= 0.0488$  | $p_y= -2.2104$ | $p_z= 0.0000$  |
| $P'$         | $E= 2.1776$ | $p_x= 0.0000$  | $p_y= 1.9652$  | $p_z= 0.0000$  |
| $q_{\gamma}$ | $E= 0.2500$ | $p_x= -0.0488$ | $p_y= 0.2452$  | $p_z= 0.0000$  |

Transverse plane

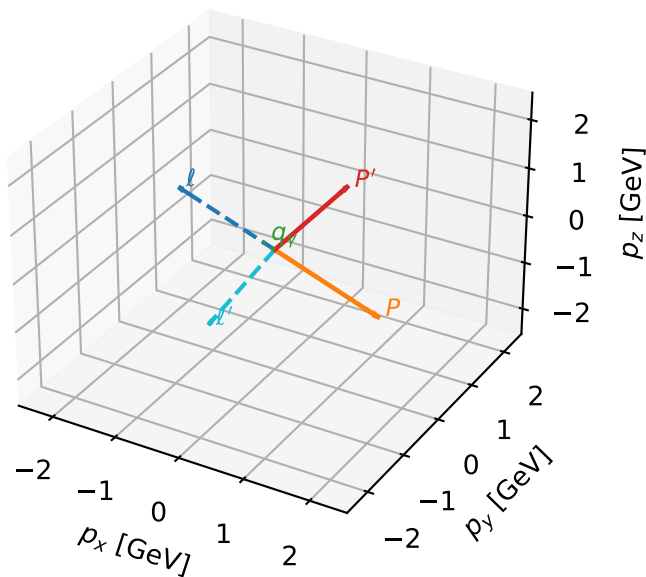


Leading initial-ensemble/final-state components (N=2, retained=99.9%)



# L massless lepton + Ty proton characteristic max $C_{\ell_0 p}$ configuration

3D momenta



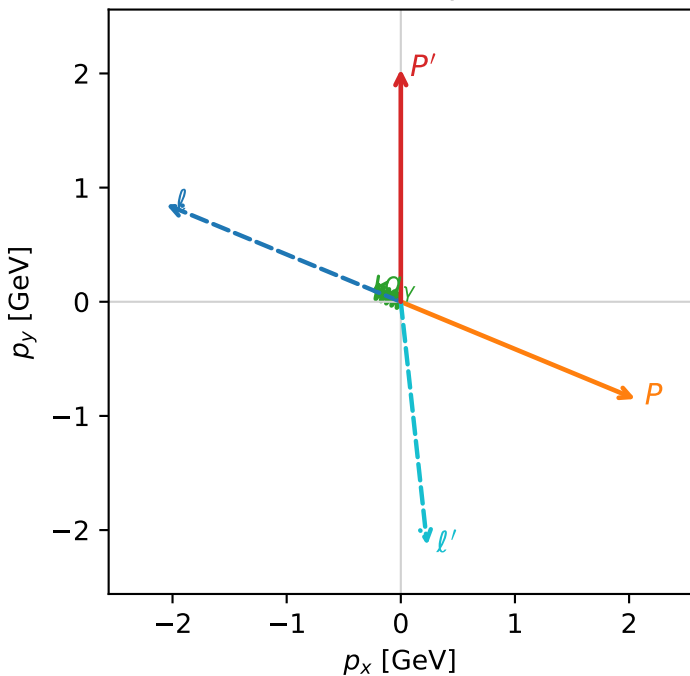
c\_ep\_low\_egamma\_lty\_region\_1 C\_{\ell\_0 p}=1.70206e-10  
 region: low\_Egamma\_LTy\_region\_1 final-pair delta\_xy=3.03373 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_{t_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.03741$   
 $\theta_{in}=1.57079$ ,  $\phi_l=2.74889$ ,  $\phi_p=5.89049$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=2.74889$   
 $Q^2=14.1261$ ,  $x_B=0.950757$ ,  $t=-12.5385$ ,  $W^2=1.61148$ ,  $y=0.719991$   
 $\theta(\ell', \gamma)=118.68$  deg

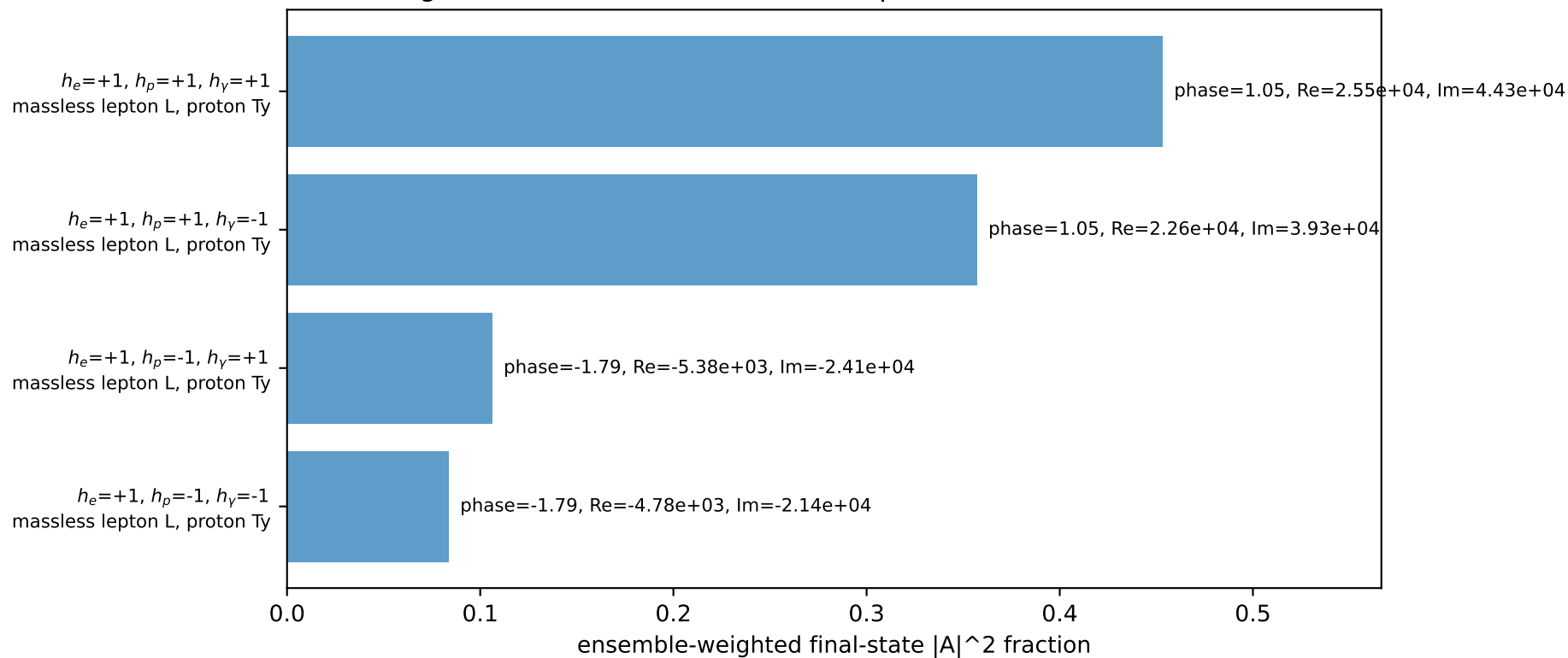
four-momenta  $[E, p_x, p_y, p_z]$  GeV:

$\ell$   $E= 2.2244$   $p_x= -2.0551$   $p_y= 0.8512$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 2.0551$   $p_y= -0.8512$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1456$   $p_x= 0.2310$   $p_y= -2.1331$   $p_z= 0.0000$   
 $P'$   $E= 2.2430$   $p_x= 0.0000$   $p_y= 2.0374$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= -0.2310$   $p_y= 0.0957$   $p_z= 0.0000$

Transverse plane

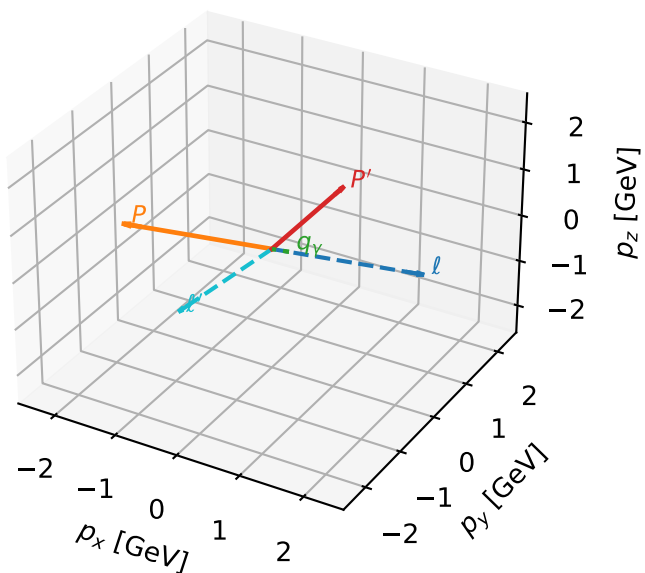


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



# L massless lepton + Ty proton characteristic max $C_{\ell_0 p}$ configuration

3D momenta

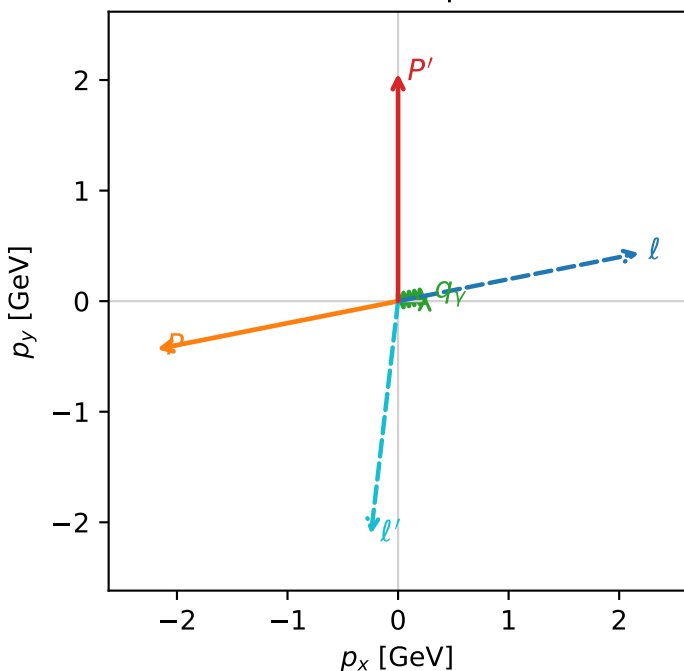


c\_ep\_low\_egamma\_lty\_region\_2  $C_{\ell_0 p}=1.31787e-10$   
 region: low\_Egamma\_LTy\_region\_2 final-pair delta\_xy=3.0259 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

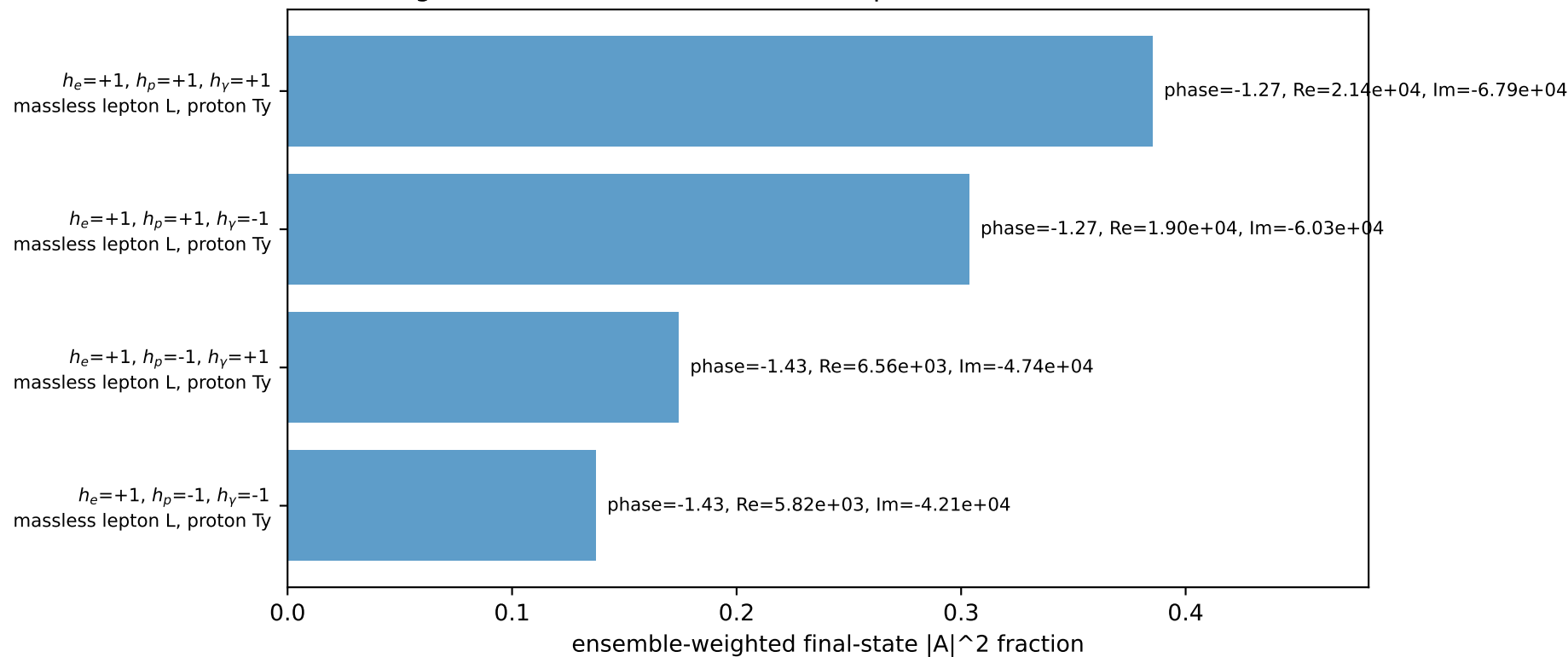
$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.06107$   
 $\theta_{in}=1.57079$ ,  $\phi_\ell=0.19635$ ,  $\phi_P=3.33794$   
 $E_\gamma=0.25$ ,  $\phi_\gamma=0.19635$   
 $Q^2=12.3506$ ,  $x_B=0.92989$ ,  $t=-10.9625$ ,  $W^2=1.81103$ ,  $y=0.643621$   
 $\theta(\ell', \gamma)=107.879$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= 2.1817$   $p_y= 0.4340$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= -2.1817$   $p_y= -0.4340$   $p_z= 0.0000$   
 $\ell'$   $E= 2.1240$   $p_x= -0.2452$   $p_y= -2.1098$   $p_z= 0.0000$   
 $P'$   $E= 2.2645$   $p_x= 0.0000$   $p_y= 2.0611$   $p_z= 0.0000$   
 $q_\gamma$   $E= 0.2500$   $p_x= 0.2452$   $p_y= 0.0488$   $p_z= 0.0000$

Transverse plane

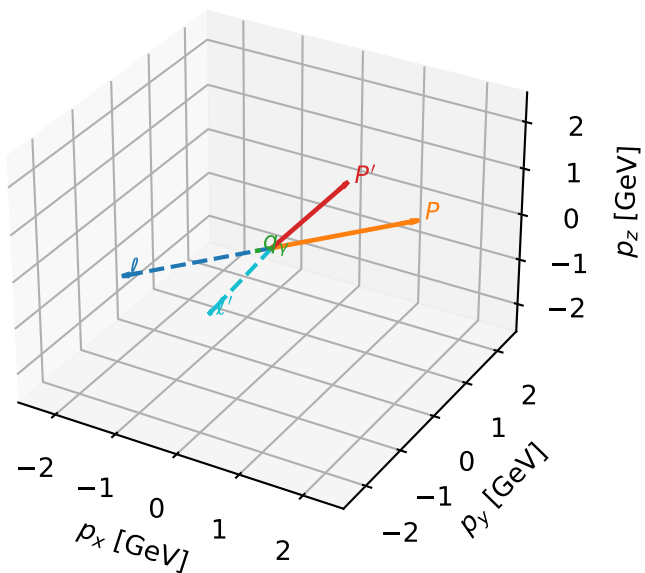


Leading initial-ensemble/final-state components (N=4, retained=100.0%)



# L massless lepton + Ty proton characteristic max $C_{\ell_0 p}$ configuration

3D momenta

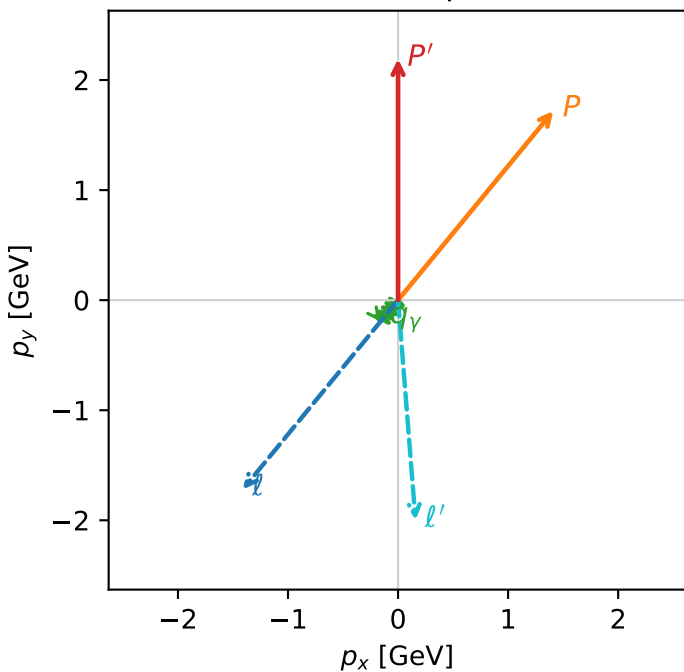


c\_ep\_low\_egamma\_lty\_region\_3  $C_{\ell_0 p}=5.46328e-11$   
 region: low\_Egamma\_LTy\_region\_3 final-pair delta\_xy=3.06239 rad  
 outgoing-state purity: 1  
 kinematic point: high\_s\_high\_theta\_in\_low\_Egamma  
 incoming state:  $|L_{\ell_0}\rangle \otimes |T_{y,p}\rangle$

$s=21.5158$ ,  $\sqrt{s}=4.63852$ ,  $|\vec{P}|=2.22442$ ,  $|\vec{P}'|=2.1916$   
 $\theta_{in}=1.57079$ ,  $\phi_{\ell}=4.02517$ ,  $\phi_P=0.883573$   
 $E_{\gamma}=0.25$ ,  $\phi_{\gamma}=4.02517$   
 $Q^2=2.49357$ ,  $x_B=0.550144$ ,  $t=-2.21332$ ,  $W^2=2.91885$ ,  $y=0.219644$   
 $\theta(\ell', \gamma)=43.9128$  deg

four-momenta  $[E, p_x, p_y, p_z]$  GeV:  
 $\ell$   $E= 2.2244$   $p_x= -1.4112$   $p_y= -1.7195$   $p_z= -0.0000$   
 $P$   $E= 2.4141$   $p_x= 1.4112$   $p_y= 1.7195$   $p_z= 0.0000$   
 $\ell'$   $E= 2.0046$   $p_x= 0.1586$   $p_y= -1.9983$   $p_z= 0.0000$   
 $P'$   $E= 2.3839$   $p_x= 0.0000$   $p_y= 2.1916$   $p_z= 0.0000$   
 $q_{\gamma}$   $E= 0.2500$   $p_x= -0.1586$   $p_y= -0.1933$   $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=100.0%)

